

Engineering Ethics Study Notes



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Chapter 2: Professionalism and Code of Ethics

Source: *Engineering Ethics* by Charles B. Fleddermann

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Chapter Goals

- Determine whether engineering is a profession.
- Understand what codes of ethics are and examine some codes from professional engineering societies.
- Understand the importance of being sensitive to ethical issues you will deal with as an engineer.



Case Study: Intel Pentium Debacle

- In **1994**, reports appeared in media that Pentium microprocessors were flawed.
- Reports published in computer journals, the *New York Times*, and daily newspapers.
- Intel first denied the problem. Later admitted the flaw but said it only affected sophisticated applications.
- Many customers requested replacement chips.
- Intel knew about the flaw and worked on a solution but initially refused replacements.
- Due to public outrage, Intel finally agreed to replace defective chips on request.

Key Question:

👉 *Did Intel do anything unethical?*

Point:

Engineers need frameworks (like codes of ethics) to understand and solve ethical problems.

Resources for Ethical Decisions

- Hallmark of professions: **Codes of Ethics** by professional societies.
 - These codes guide professionals when facing ethical problems.
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Is Engineering a Profession?

Definitions

- **Job:** Any work for hire.
- **Occupation:** Employment through which someone earns a living.
- **Profession:**
 - Requires sophisticated skills, judgment, discretion.
 - Needs extensive formal education.
 - Controlled by societies that set standards for admission.
 - Aims to achieve public good.

Examples:

- *Professional athlete* → distinguishes from amateur.
 - *Professional carpenter* → skilled with years of experience.
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Attributes of a Profession

1. **Sophisticated Skills:** Involves judgment and discretion.
2. **Formal Education:** Requires extensive study.
3. **Professional Societies:** Set and maintain standards.
4. **Public Good:** Serves society's benefit.

Judgment:

Not simple day-to-day decisions, but significant ones with large impact (lives, money).

Discretion:

- A) Confidentiality — keeping client/employer information private.
 B) Moral autonomy — freedom to make independent decisions.
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Engineering Qualifies

- Involves sophisticated skills and judgment (e.g., design decisions).
- Discretion: protecting intellectual property.
- Core concern: public safety.
- Note: Unlike medicine/law, most engineers are **not self-employed**.

Difference from Other Professions

- Law & Medicine require licenses; many engineers do not.
- If engineering were like medicine:
 1. 4-year “pre-engineering” (math, physics, CS).
 2. 3–4-year “Doctor of Engineering” professional degree.

Models of a Profession

1. **Social Contract Model:** Serve public good.
2. **Business Model:** Serve members’ economic advantage.

 **Engineering = Social Contract Model**

Codes of Ethics

- Express rights, duties, and obligations of members.
- Provide a framework for making ethical decisions.
- Not legally binding (no arrest) but violation can lead to censure or expulsion.
- Restate existing standards; do not create new ones.
- Create a culture where ethical behavior is the norm.

Objections

- Few engineers are members of societies or know the codes.
- Codes can have **internal conflicts**.
- Codes often lack methods for resolving conflicts.

History

- Engineering societies formed in late 19th century (USA).
- Early 20th century: focused on business conduct (no advertising or competitive bidding).
- Modern codes: emphasize public safety, health, environment.

Examples: IEEE vs. NSPE

IEEE	NSPE
Short and general	Longer and detailed

IEEE	NSPE
No mention of duty to employer	Explicitly states duty to employer
Mentions duty to environment	Focus on public health

Resolving Internal Conflicts

Example:

- Boss orders unsafe design.
- NSPE:
 - Duty to employer (clause 1.4).
 - BUT public safety (clause 1.1 & preamble) overrides.
- Result: *Public safety comes first.*

Protection for Employees

- Societies help employees resist unethical pressure.
- Codes provide “ammunition” to justify resisting unethical conduct.

Corporate Codes

- Not only professional societies: universities, corporations have codes too.
- Corporate codes:
 - Raise awareness of ethical issues.
 - Promote ethical culture.
 - Cannot cover every situation.

Detailed Case: Intel Pentium Debacle

About Intel:

- US semiconductor maker, Santa Clara, CA.
- Inventor of x86 microprocessors.
- Produces motherboards, memory, graphics chips.

Problem:

- Flaw in floating point unit (FPU) caused wrong results in double precision arithmetic.

Intel's Response:

1. Denied issue.
 2. Admitted but minimized significance.
 3. Negative publicity → agreed to replace chips.
 4. Policy shift: fix problems immediately, let customers decide if significant.
 5. Cost: \$475 million.
 6. 1997: similar flaw found → promptly fixed + software patch offered.
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Other Case Studies to Review

- Runway Concrete at Denver International Airport.
 - Competitive Bidding and the Paradyne Case.
 - Application of Biomedical Engineering.
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End of Notes

If you'd like, I can:

- ✓ Make a **Word document**
- ✓ Make a **Markdown file**
- ✓ Create a **printable PDF**
- ✓ Design **flashcards** from these notes
- ✓ Or help you prepare **practice questions**

Would you like me to do any of these?  ✨



give me all practice questions but answer the in details please

Perfect — let's turn your **Chapter 2: Professionalism and Code of Ethics** content into detailed practice questions **with full, clear answers**. I'll cover all the main areas: **definitions, differences, codes, real-world cases, and conflicts**.

PRACTICE QUESTIONS AND DETAILED ANSWERS

1 What is the difference between a *job*, an *occupation*, and a *profession*? Give examples for each.

Answer:

- **Job:** Any work done for pay, regardless of the skill or responsibility involved.
Example: A part-time cashier at a grocery store.

- **Occupation:** Work that a person does to earn a living — implies longer-term employment or career field.
Example: An electrician working for a construction company.
 - **Profession:** A career that requires sophisticated skills, extensive formal education, specialized training, and involves judgment and discretion. It is regulated by professional societies and aims to serve the public good.
Example: A licensed civil engineer designing bridges, a doctor treating patients, or a lawyer practicing law.
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2 What are the four main attributes of a profession? Explain each with an example.

Answer:

1. **Sophisticated Skills:**

Professionals must apply advanced knowledge and use judgment.

Example: An aerospace engineer deciding how to balance weight and safety in an aircraft design.

2. **Formal Education:**

Requires years of structured learning.

Example: A structural engineer must complete a university engineering degree and possibly postgraduate studies.

3. **Professional Societies:**

Membership and certification standards are maintained by governing bodies.

Example: IEEE for electrical engineers, NSPE for professional engineers.

4. **Public Good:**

A profession must benefit society through its practice.

Example: A biomedical engineer designs medical devices that save lives.

3 What does “judgment” mean in the context of a profession, and how is it different from everyday decision-making?

Answer:

In a profession, **judgment** means making significant, impactful decisions using formal training and experience. These decisions often involve safety, large financial investments, or human lives.

Example:

- *Professional Judgment:* An engineer deciding whether a bridge can safely carry the projected traffic load.
 - *Everyday Judgment:* A secretary deciding which email to reply to first — important but not life-altering.
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4 What does “discretion” mean in the context of engineering? Give both meanings.

Answer:

1. Confidentiality:

The responsibility to keep sensitive information about clients, employers, or designs secret.

Example: An engineer working on a new chip design must not share details with competitors.

2. Moral Autonomy:

The freedom and responsibility to make ethical decisions independently when facing moral dilemmas.

Example: An engineer refusing to approve a design they know is unsafe, even under pressure from a boss.

5 Compare engineering with medicine and law. How is engineering similar or different as a profession?

Answer:

✓ Similarities:

- All three require sophisticated skills, judgment, and formal education.
- All serve the public good.

✗ Differences:

- Medicine and law require **licenses** for all practitioners — not all engineers must be licensed to work.
- Doctors and lawyers are often **self-employed** (run private practices), whereas most engineers work as employees for companies.
- If engineering followed the same training path, engineers might need a “Doctor of Engineering” degree after a pre-engineering undergraduate program.

6 What are the two models of professions? How does engineering fit in?

Answer:

1. Social Contract Model:

Professions primarily exist to serve the public good.

Example: Engineering — safety and welfare of the public are top priorities.

2. Business Model:

Professions aim to protect and advance the economic interests of their members.

Example: A trade association that negotiates higher pay for its members.

✓ *Engineering* best fits the **social contract model** because its main purpose is to serve society (public safety, health, environment).

7 What is a Code of Ethics? What is its purpose in engineering?

Answer:

A **Code of Ethics** is a set of principles that outlines the rights, duties, and obligations of members of a

profession.

Purpose:

- Provides a framework for ethical judgment.
- Helps professionals make consistent, responsible decisions.
- Promotes a culture where ethical behavior is the norm.
- Reinforces existing ethical principles but does not create new ones.
- Violating it is not illegal but can lead to expulsion from professional societies.

Example: IEEE and NSPE have Codes of Ethics for engineers.

8 What are some objections to engineering codes of ethics?

Answer:

- Many practicing engineers are not members of professional societies, so they may not follow the codes.
 - Even members often do not know or read the codes.
 - Codes can contain **internal conflicts** (e.g., duty to employer vs. duty to public safety).
 - Codes rarely explain how to resolve conflicts when principles clash.
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9 Explain how an internal conflict might occur in a code of ethics. Use the NSPE example.

Answer:

Example Scenario:

- An engineer's boss instructs them to approve a design they know is unsafe.
- NSPE Code of Ethics says engineers have a **duty to their employer** (clause 1.4).
- But the code's **preamble and clause 1.1** make clear that public safety comes first.

Resolution:

When principles clash, the **higher priority is public safety** — so the engineer must refuse to approve the unsafe design and try to convince the employer to change it.

10 How can professional societies and codes protect employees who face unethical pressure?

Answer:

Professional societies provide a **backbone** for engineers pressured to act unethically.

- An engineer can point to the Code of Ethics to justify refusing unsafe or dishonest actions.
- If punished or threatened, they can use the society's support as "ammunition" to resist or blow the whistle on unethical conduct.

1 1 What other organizations, besides professional societies, might have codes of ethics?

Answer:

- **Corporations:** Many companies have their own codes to raise ethical awareness and promote an ethical culture internally.
- **Universities:** Student codes of conduct or honor codes.
- **Special Areas:** For example, codes for ethical computer use, research integrity, or AI ethics.

1 2 Explain the Intel Pentium debacle. What happened? Why is it an ethical case study?

Answer:

Background:

- Intel, a major chip manufacturer, produced Pentium microprocessors in 1994.
- There was a flaw in the chip's **floating point unit** that caused errors in double precision arithmetic.
- Intel first **denied** the problem, then minimized its importance.
- Public backlash forced Intel to offer replacements.
- The incident cost Intel **\$475 million**.
- In 1997, a similar flaw was handled transparently and fixed quickly.

Ethical Lesson:

- Companies should be transparent and proactive about product flaws, especially when they affect customers.
- Engineers must push for honesty and safety even when corporate interests conflict.
- The case shows why codes of ethics and frameworks are necessary to judge what is right.

✓ End of Detailed Practice Q&A

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- Generate **flashcards** 📇
- Create **scenario-based case study questions** 🧩
- Or make a **printable PDF** for your exam prep!

Would you like me to package this up for you? 🎓 ✨



give me a table of IEEE vs. NSPE

Absolutely! Here's a clear **comparison table** for **IEEE vs. NSPE Codes of Ethics**, based on your extracted PDF content:

 **Table: IEEE vs. NSPE Codes of Ethics**

Aspect	IEEE Code	NSPE Code
Length & Style	Short and general	Longer and more detailed
Structure	General principles	Has a preamble that summarizes the engineer's duties, then goes into explicit clauses
Duty to Employer	Does <i>not</i> mention a specific duty to the employer	Explicitly mentions duty to the employer
Duty to Public Health & Safety	Mentions duty to protect the environment	Emphasizes duty to protect public health and safety
Environment	Explicitly mentions environmental protection	Focuses more on public safety, less explicit about the environment
Conflict Resolution	Less detailed guidance for resolving conflicts	Provides clauses but conflicts can arise (e.g., clause 1.4 duty to employer vs. clause 1.1 duty to public safety)
Focus	Broad and principle-based	More prescriptive, outlines detailed responsibilities

✓ Key Takeaway

- **IEEE** is broader and more principle-based — good for general guidance.
- **NSPE** is more specific — useful for clear rules on handling duties to the public, employer, and conflicts.

If you want, I can turn this into:

- ✓ A printable **PDF table**
- ✓ A **flashcard** version
- ✓ Or add **real-world scenarios** to show how to apply each code.

Would you like me to prepare any of these?  ✨